



SCHOOL OF COMPUTER ENGINEERING
KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY (KIIT)
(Deemed to be University, u/s 3 of UGC Act 1956)

Subject: Network and Cyber Security (7th Semester)

Subject Code: CS40017

Academic Session: Autumn Semester 2026

Course Teacher: Prof.(Dr.) Prachet Bhuyan

Contact hours per week: 3 hours (LTP: 3-0-0)

Course Objectives:

- To gain knowledge about the cybersecurity in the real world applications.
- To understand the requirements of security protocols in the networks.
- To learn about the ethics and cyber laws.
- To realize different authentication mechanisms.

Course Contents:

UNIT I

Introduction to Network Security: Introduction to Network security, Model for Network security, Model for Network access security, Real-time Communication Security: Introduction to TCP/ IP protocol stack, Implementation layers for security protocols and implications, IPsec: AH and ESP, IPsec: IKE.

UNIT II

Cyber Security: Introduction to cyber security, Media-Based-Vulnerabilities, Network Device Vulnerabilities, Back Doors, Denial of Service (DoS), Spoofing, Man-in-the-Middle, and replay, Protocol-Based Attacks, DNS Attack, DNS Spoofing, DNS Poisoning, ARP Poisoning, TCP/ IP Hijacking, Virtual LAN (VLAN), Demilitarization Zone (DMZ) , Network Access Control (NAC), Proxy Server , Honey Pot , Network Intrusion Detection Systems (NIDS) and HostNetwork Intrusion Prevention Systems Protocol Analyzers, Internet Content Filters

UNIT III

Cyber Law and Ethics: Policy vs. Law, Types of Law, General Computer crime laws, U.S. copyright law, U.K. computer security laws, Ethics and education, Codes of ethics of professional organizations.

UNIT IV

Authentication: Kerberos, X.509 Authentication Service, Scanning: Port Scanning, Port Knocking- Advantages, Disadvantages. Peer to Peer security. Electronic Mail Security: Distribution lists, Establishing keys, Privacy, source authentication, message integrity, non-repudiation, proof of submission, proof of delivery, message flow confidentiality, anonymity, Pretty Good Privacy (PGP)

UNIT V

Firewalls and Web Security: Packet filters, Application level gateways, Encrypted tunnels, Cookies.

Course Outcomes:

Upon completion of this course, the students will be able to:

CO 1: Distinguish and analyze available network and network layer security such as IPSec

C02: Analyze and evaluate the cyber security needs of an organization

C03: Access various threats and vulnerabilities of a network, and explain appropriate countermeasures

C04: Illustrate the legal, ethical and professional issues in cyber security

C05: Access the authentication in networks and secure mail services

C06: Justify the usage of firewalls and gateways

Course Coverage and Delivery Plan:

Unit Name	Topics/Coverage	No. of lecture
Introduction to Network Security	• Introduction to Network security, • Model for Network security, • Model for Network access security, • Real-time Communication Security: ➤ Introduction to TCP/ IP protocol stack, ➤ Implementation layers for security protocols and implications, ➤ Diffie-Hellman (DH) key exchange ➤ IPsec: AH and ESP, IPsec: IKE	6
Cyber Security	• Introduction to cyber security, • Media-Based-Vulnerabilities, • Network Device Vulnerabilities, • Back Doors, • Denial of Service (DoS), • Spoofing, • Man-in-the-Middle, and replay, Protocol-Based Attacks, • DNS Attack, DNS Spoofing, • DNS Poisoning, ARP Poisoning, • TCP/ IP Hijacking, Virtual LAN (VLAN), • Demilitarization Zone (DMZ) , Network Access Control (NAC), Proxy Server ,Honey Pot , Network Intrusion Detection Systems (NIDS) Host Network Intrusion Prevention Systems Protocol Analyzers, Internet Content Filters	12
MID SEMESTER		
Cyber Law and Ethics	• Policy vs. Law, • Types of Law, • General Computer crime laws, • U.S. copyright law, • U.K. computer security laws, • Ethics and education, • Codes of ethics of professional organizations.	4

Authentication	• Kerberos, • X.509 Authentication Service, • Scanning: Port Scanning, • Port Knocking- Advantages, Disadvantages. Peer to Peer security. Electronic Mail Security:- • Distribution lists, • Establishing keys, • Privacy, • source authentication, • message integrity, • non-repudiation, • proof of submission, • proof of delivery, • message flow confidentiality, • anonymity, • Pretty Good Privacy (PGP) –	10
Firewallsand Web Security	• Packet filters, • Application level gateways, • Encrypted tunnels, • Cookies.	4

Text books:

TB1. William Stallings, "Cryptography and Network Security Principles and practice", 8th Edition, Pearson Education, 2023.

TB2. Michael E. Whitman, Herbert J Mattord, "Principles of Information Security", Cengage, Seventh Edition, 2023.

Reference Books:

R1. Atul Kahate, "Cryptography and Network Security", 4th edition, McGraw Hill, 2019.

R2. Bernard L. Menezes, Ravinder Kumar, "Cryptography, Network Security and Cyber Laws", 1st edition, cengage, 2018.

R3. Charlie Kaufman, Radia Perlman, Mike Speciner, Ray Perlner, "Network Security: Private Communications in a Public World", 3rd edition, Pearson, 2024.

R4. Cryptography And Network Security (SIE) 2ND Edition by Behrouz A. Forouzan and Debdeep Mukhopandhyay

Scheme of Evaluation:

Full marks for the Network and Cyber Security theory is 100, which is divided into the following components.

- Internal Assessment (30 Marks)
- Mid Semester (20 Marks)
- End Semester (50 Marks)

Activity Details and Tentative Schedule:

Activity	Type	Focus	Date	Marks	CO
1	Assignment-1	Critical Thinking	4 th week of July, 2025	5	CO1
2	Assignment-2	Critical Thinking	2 nd week of August, 2025	5	CO2
3	Quiz-1	Quiz	4 th week of August, 2025	5	CO3
MID SEMESTER Examination [08-13 September 2025]					
4	Assignment -3	Critical Thinking	4 th week of September, 2025	5	CO4
5	Group Activity & Presentation	Creation, Reflections	2 nd week of October, 2025	5	CO5
6	Quiz-2	Quiz	4 th week of October, 2025	5	CO6
END SEMESTER Examination [10-19 November 2025]					

Note: Defaulter Tests may be conducted for the genuine absentees.